Teaching a student who forgets: one-page summary

Thesis work on adaptive tutoring (cursim). Results of 2026-10-18.

Question

Students forget during breaks. **If the break length is known, how should a tutor change what it does?** We vary two settings: **revision budget** (share of questions spent on topics already learned) and **study depth** (right answers in a row needed before moving on).

Setup

A simulated student (40 topics with prerequisites) practises 20 days, 25 questions a day, then takes a 0 to 28 day break. We count topics still usable afterwards (maximum 40), for memory half-lives of 7, 14 and 28 days. Best settings are picked on 50 students and scored on 100 new ones.

Tutors

CurriculumTutor, coded exactly as in the paper (our earlier version was a stand-in, so old comparisons are retired), plus one or two settings changed. Revision 0 and depth 7 is the paper's tutor. Adaptive review is dropped.

Findings (14-day half-life)

- **The published tutor is far from the best:** 15.4 topics after a 3-day break against 33.6 for the best settings, 17.4 topics short on average.
- **Longer break, more revision:** best budget 0.1 (no break), 0.3 (3 days), 0.4 (7), 0.5 (10).
- **Depth alone does nothing;** with revision, shallow depth (about 3 in a row) is best, and it deepens only for long breaks (8 at 10 days).
- **No single setting wins everywhere:** the best fixed pair still loses 3.4 topics on average, 10.3 at worst.

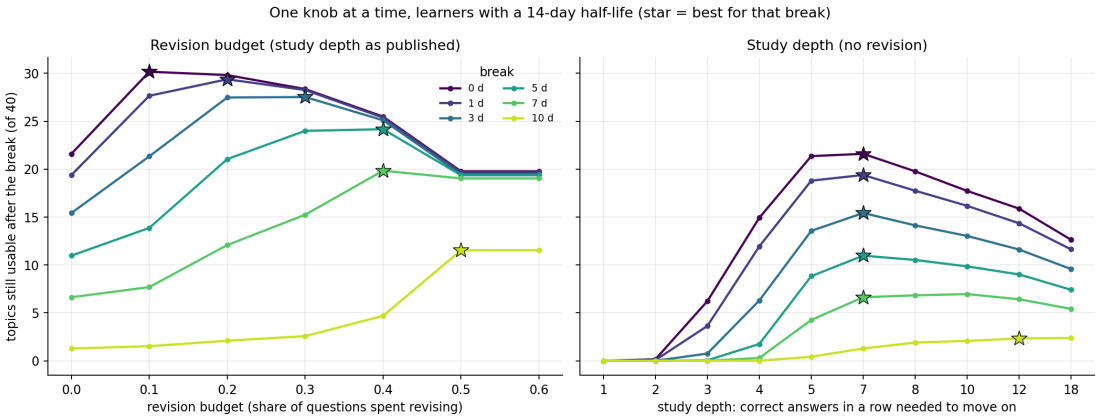


Fig. 1. Each knob alone, one line per break length; stars mark the best setting.

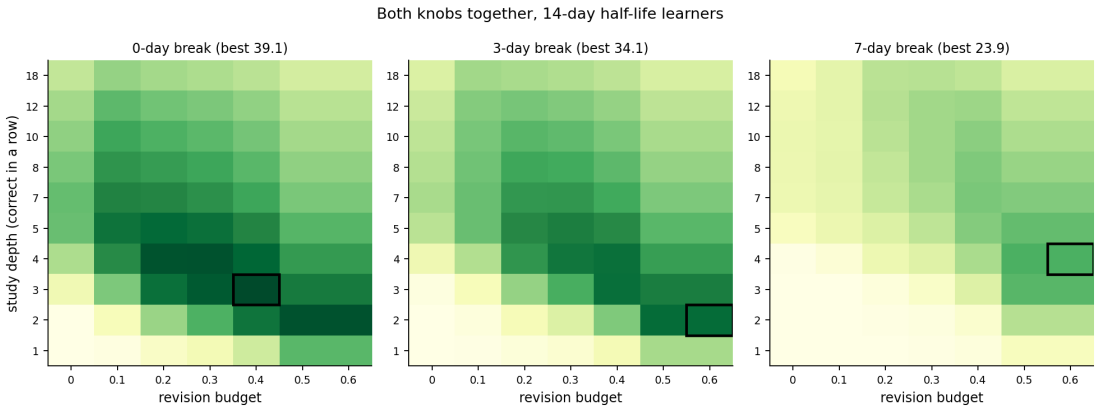


Fig. 2. Both knobs together (darker = more topics usable); the box marks the best cell.

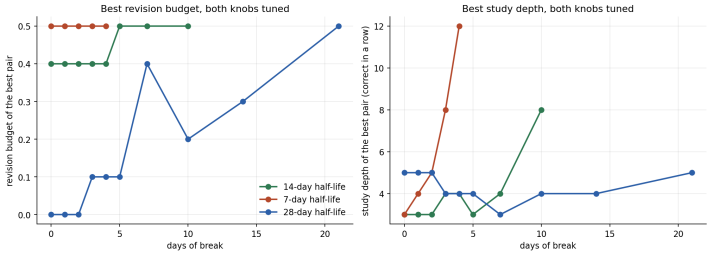


Fig. 3. Best pair against break length for three memory speeds (noisy where few topics remain).

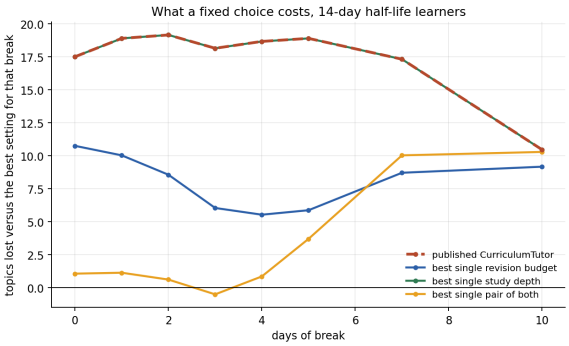


Fig. 4. Topics lost by fixed settings; the published tutor and depth-only coincide.

Limits: simulated students and a made-up curriculum; best settings are slightly flattered by being chosen on a small group.
Next: a tutor that sets both knobs without being told the break, then real learner logs. Details: experiment_logs.md.